

PIXE Analysis on Maya Blue in Prehispanic and Colonial Mural Paintings

Group_Project_8_Presentation

Awet, Connor, Lachlan, and Pablo

Introduction

Background

- ▶ **Maya Blue:** a distinctive blue-green pigment used in Prehispanic and colonial murals across southern Mexico
- ▶ **PIXE** (Particle-Induced X-ray Emission): measures elemental concentrations non-destructively
- ▶ **Dataset:** 35 mural samples, 17 elements (Na–Pb), from three archaeological groups:
 - ▶ *CEC* — Santa Cecilia Acatitlán (post-classic, non-fresco)
 - ▶ *Prehispanic* — Cacaxtla, Tajín, Tamuin
 - ▶ *Convents* — Mexican Franciscan colonial sites



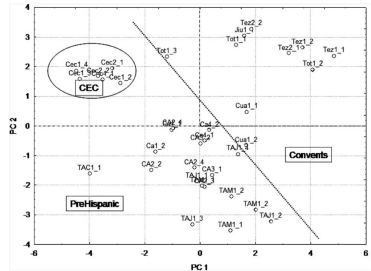
Questions to Answer

There are **4 main questions** that we set out to answer during this project

- ▶ Are there any surprising observations in the data?
- ▶ Can we assume that the data follows a multivariate normal distribution?
- ▶ Can we find groups in this data?
- ▶ Can we recreate figure 2 from the original study?

Figure 2. from the original Paper

- ▶ **Original Figure** shows clustering of 3 types of Muarls
 - ▶ *CEC* Cluster in the top left; Very compact together
 - ▶ *PreHispanic* Cluster across the bottom; spread through left to middle
 - ▶ *Convents* Cluster in top right; Cua Moving closer to PreHispanic



Method

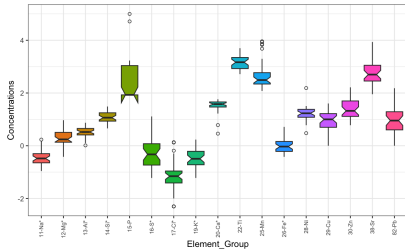
Data Transformation

Original Data vs Transform Data

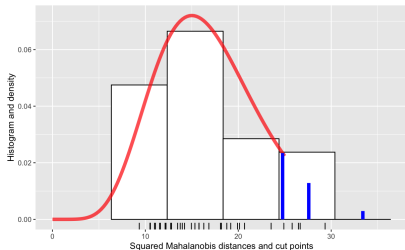
Descriptive Statistics

Descriptive Statistics of Elemental Concentrations								
Log10 Transformation								
Element	Mean	Standard D.	Median	Minimum	1st Quartile	Inter-Q Range	3rd Quartile	Maximum
11-Na*	-0.44	0.28	-0.48	-0.96	-0.65	0.35	-0.30	0.23
12-Mg*	0.33	0.35	0.24	-0.42	0.13	0.37	0.51	0.97
13-Al*	0.50	0.19	0.52	0.01	0.41	0.24	0.65	0.87
14-Si*	1.09	0.22	1.06	0.66	0.93	0.32	1.25	1.49
16-S*	-0.23	0.61	-0.33	-1.22	-0.73	0.81	0.08	1.11
17-Cl*	-1.14	0.63	-1.15	-2.30	-1.41	0.45	-0.96	0.13
19-K*	-0.47	0.35	-0.49	-1.22	-0.72	0.51	-0.21	0.23
20-Ca*	1.50	0.26	1.58	0.46	1.47	0.19	1.66	1.76
26-Fe*	-0.02	0.25	-0.03	-0.42	-0.21	0.38	0.16	0.71
15-P	2.53	0.78	1.93	1.93	1.93	1.11	3.04	5.00
22-Ti	3.18	0.28	3.17	2.72	2.93	0.42	3.35	3.70
25-Mn	2.67	0.55	2.49	2.08	2.34	0.43	2.77	3.96
28-Ni	1.23	0.27	1.23	0.48	1.08	0.30	1.38	2.19
29-Cu	0.95	0.38	1.00	0.00	0.70	0.52	1.22	1.60
30-Zn	1.38	0.37	1.32	0.78	1.11	0.59	1.70	2.21
38-Sr	2.74	0.45	2.70	1.95	2.45	0.60	3.05	3.93
82-Pb	0.96	0.61	0.95	0.00	0.60	0.69	1.29	2.18

Qualitative Analysis



Are There Surprising Observations?



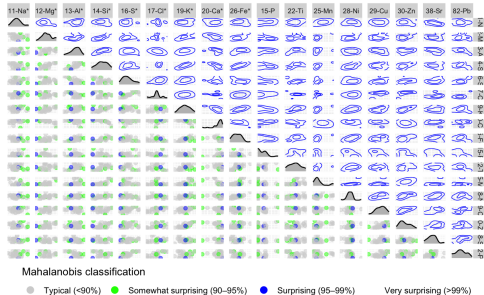
- ▶ The 90%, 95%, and 99% Cut off values were: 24.76904, 27.58711, and 33.40866 Respectively
- ▶ We Identified 1 Suprising and 4 somewhat suprising observations
- ▶ TAJ1_4 was Identified as the suprising observation; $DM^2 = 29.34327$

Closer Look at Suprising Observations

Sample	dM	surprise
TAJ1_4	29.34327	Suprising
Tot1_1	26.66402	somewhat
TAJ1_1	26.47125	somewhat
Jiu1_1	25.79784	somewhat
Cec1_3	24.90653	somewhat

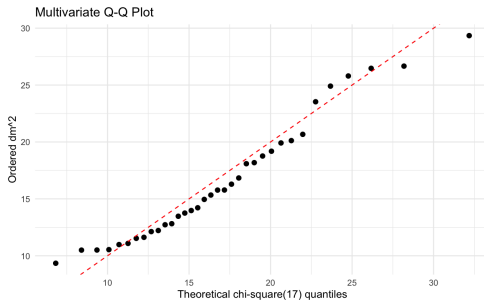
The largest Mahalanobis distances identify samples with unusual combinations of elemental concentrations. TAJ1_4 is the most extreme observation, suggesting that its overall elemental profile differs most strongly from the centre of the dataset.

Pairs Plot of Mahalanobis Distances and Contour Densities



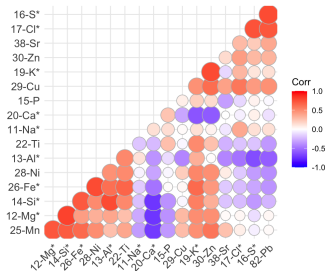
- ▶ Each panel compares two elements pairwise
- ▶ Contours show where observations are most concentrated
- ▶ Highlighted points flag unusual multivariate behaviour
- ▶ Outliers are rarely extreme in a single element — their unusualness is multivariate

Can we assume that the data follows a multivariate normal distribution?



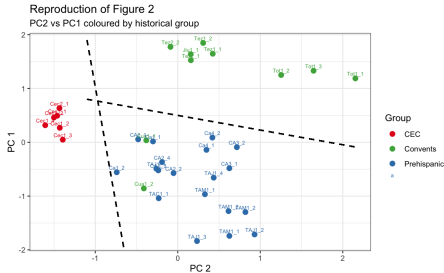
- ▶ Fails Mardia's skewness test ($p = 0.003$) — **not multivariate normal**
- ▶ Kurtosis result is consistent with normality

Correlogram



- ▶ Red = positive correlation, blue = negative; larger circles indicate stronger relationships
- ▶ Mn, Mg, Si, Fe, Ni, Al, Ti, K, and Zn are mutually positively correlated
- ▶ Na, Ca, P, S, and Pb are negatively or uncorrelated with that group

Can we find groups in this data?



- ▶ **CEC** forms a tight, well-separated cluster
- ▶ **Prehispanic** and **Convents** overlap substantially
- ▶ Some structure exists, but groups are not fully distinct

Discussion

The data also showed some surprising observations. TAJ1_4 had the biggest Mahalanobis distance and was classified as surprising, while Tot1_1, TAJ1_1, Jiu1_1 and Cec1_3 were all classified as somewhat surprising. The clustering analysis provides evidence of some meaningful groups in the data, with the CEC samples forming a distinct group. This is consistent with the authors' observation of these samples containing unusually low calcium concentrations and likely not being produced with the fresco technique. The samples also show a less clear distinction between predominantly Prehispanic and Colonial samples, as these two groups have substantial overlap rather than being completely separated.

Discussion

We agree with the authors' conclusions about the zones in Figure 2. The original PCA differentiated samples into broad Prehispanic and Colonial patterns and identified CEC as a distinct cluster. They also concluded that the data did not form well-defined groups. We were able to reproduce a similar graph, providing evidence of these findings. Elemental composition provides information about the historical period and the technique used, but the overlap between samples means that the groups cannot be treated as completely separated categories. The authors' hypothesise that the overlap may be due to similarities in the painting techniques between the Prehispanic and Colonial periods.

Conclusion

Overall, the results show some structure in the Maya blue samples, but the groups are not completely distinct. The CEC samples are clearly separated, while the Prehispanic and Colonial samples overlap. Figure 2 supports the authors' conclusion that the zones represent broad patterns rather than clearly defined groups.

References

Sánchez del Río, M., Martinetto, P., Solís, C., & Reyes-Valerio, C. (2006). *PIXE analysis on Maya blue in Prehispanic and colonial mural paintings*. Nuclear Instruments and Methods in Physics Research Section B, 249, 628–632.